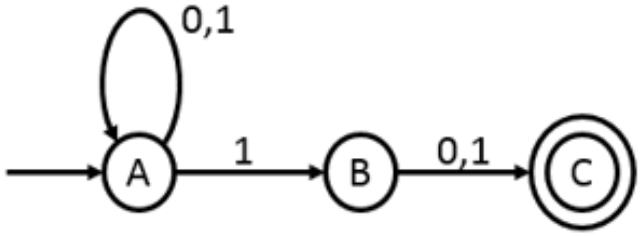
1. What is alphabet, word, Language, grammar in automata theory?
2. Give an example for finite language and an example of infinite language?
3. If $\sum = \{0,1\}$, what is $\sum^2$ and what is $\sum^*$?
4. State different proof techniques?
5. Define DFA?
6. Preparing Transition function for any given DFA?
7. Constructing DFA from a given transition function?
8. Construct DFA for the below languages:
    a. Construct DFA which accepts the set of strings {1101, 101, 111}?
    b. All strings containing 01 as substring?
    c. All string ending with 01?
    d. All string containing 11 as substring?
    e. All string starting with a 1 and ending with a 1?
    f. Binary string that has even number of 0s and even number of 1s?
    g. All the binary strings that have 110 as substring?
    h. Binary strings whose decimal equivalent is divisible by 3?
    i. Binary string whose decimal equivalent is divisible by 5?
    j. Binary string that starts with 10 and ends with 11?
    k. All the strings beginning with 101?
    l. All the strings containing 1101as substring?
    m. Set of all string which exactly three consecutive zeros?
    n. Construct a DFA for the set all strings over {a, b} with odd number of b's?
    o. Construct a DFA for the set of all strings containing 1011 as substring?
    p. Construct a DFA for the set of all strings beginning with a 1 that, when interpreted as a binary integer, is multiple of 5. For example, strings 101, 1010 and 1111 are in language; 0, 100, and 111 are not in the language?
9. Define NFA and how is it different from DFA?
10. When does a word is accepted by an NFA or an e-NFA?
11. Construct an NFA for the following languages:
    a. Construct an NFA that accepts all strings over {a, b} that end with a?
    b. Construct an NFA over {0,1} that accepts strings with at least one 0 and at least one 1?
    c. Construct an NFA that accepts strings ending with 01 or 10?
    d. Construct an NFA for the set of all strings containing either 101 or 110 as substring?
    e. Construct an NFA for the set of all strings over {a,b} that starts and ends with the same letter?
    f. Construct an NFA for the set of all strings such that every 1 is followed immediately by 00?
    g.
12. Converting NFA to DFA using subset construction method?
13. Define is ε-NFA, how is it different from DFA/NFA?
14. What is ε-Closure of a state?
15. Consider the following ε-NFA:

δ	a	b	c	ε
-> p	Ø	{q}	{r}	{q, r]
q	{p}	{r}	{p, q}	Ø
*r	Ø	Ø	Ø	Ø

A. Compute the ε-closure of each state
B. Give the set of all strings of length 3 or less accepted by the automaton
C. Covert the automaton to a DFA

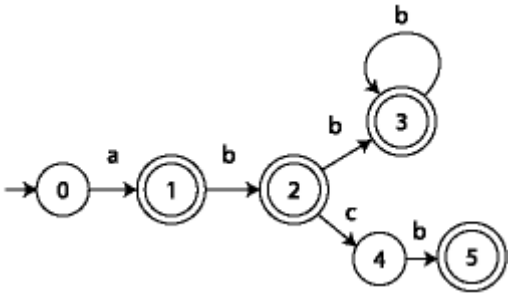
16. Design ε-NFA for the set of strings such that at least one of the last 3 positions is a 1?
17. Cover the following NFA to DFA?



18. Write Regular Expressions for the following lnaguages:
    a. The set of all strings that begin with 110
    b. The set of all strings that contain 1011
    c. The set of all strings that contain exactly three 1's
    d. The set of all strings such that the number of 0's is odd
    e. The set of all strings that do not contain 101 as substring
    f. The set of all strings over {a,b} that do not contain bb?
    g. The set of all strings over {a,b,c} that start and end with the same symbol?
    h. The set of all strings over {a,b} where every a is immediately followed by b?
    i. The set of all strings over {0,1} whose length is a multiple of 3?
19. State the pumping lemma for regular languages?
20. Prove that the language L = {ww | w ∈ {a,b}*} is not regular by using pumping lemma?
21. Prove that the language L = {$a^i b^j$ | i > j } is not regular by using pumping lemma?
22. Construct a DFA for $L_1 \cap L_2$, where

$$L_1 = \{w \in \{0,1\}^* \mid w \text{ ends with } 1\}$$
$$L_2 = \{w \in \{0,1\}^* \mid w \text{ has even number of 0s}\}$$

23. Compute regular expression for the below NFA?



24. Conversion of regular expression to ε-NFA?

25. State 5 words that can be derived from the regular expression: $(0^*1^*)^*000(0+1)^*$?
26. Conversion of DFA to regular expression by using state elimination method?
27. Convert the following regular expressions to ε-NFA:
    a. 01 + 101
    b. $(01 + 1)^*$
    c. $011\ (0 + 1)^*$
28. State any two applications of regular expressions?
29. Define Context Free Grammar (CFG)?
30. Deriving a word from CFG, left most derivation and right most derivation?
31. Construction of parse tree for a word of a CFG?
32. The following grammar generates the grammar of the language consists of strings of even length: S -> SS| aa| ab | ba | bb | ε
    a. Give left most derivative for the string aabbba
    b. Give right most derivative for the string baabab
    c. Give any parse tree for the string aaabbb
33. State any two applications of Context Free Grammars?
34. Define push down automata?
35.